

Opera Numerorum

Machine Certification for $\text{GRH}(X_{143})$ and $\text{BSD}(J_{143})$

David J. Fox

ORCID: 0009-0008-1290-6105 | davidjfox998@gmail.com | Aberdeen/Seattle, WA

June 4, 2026

CERTIFIED: $\text{GRH for } X_{143} \bullet \text{BSD rank}(J_{143}) = 1 \bullet \alpha_{143} = 299 + \pi/10$

Master Manifest SHA-256 (M1–M6 concatenated stdout)

5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9

SHA-256 Certified Chain

Mod	Claim	Stdout SHA-256	Status
M1	$\alpha_0 = 299 + \pi/10$ (5000 dps)	63ef870a78766619327e99b68683bceff8c8ef9a525298756c77c8378fd2c291	CERTIFIED
M2	kappa bound (80-bit long double)	3716c7dbb32524074b8fffb65eea45069c8b568a31dc73706405116b84029a83	CERTIFIED
M3	CF $\pi/10$: $Q_5=226$, bound=82829	e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044	CERTIFIED
M4	S_{14} : 14 primes, $p_5 > 82829$	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed	CERTIFIED
M5	$C(S_4) = 11.4221 > 2 \cdot \sqrt{13}$	9df98a3970acbb6942770a6cdd42fb21b009f9a5f45a222dd963e98ba4cb7a13	CERTIFIED
M6	$\text{genus}(X_0(143))=13$, Bost bound	ec9fa8c3aad478312c7e0d7373904dc3407eb5e9f4c19a011e3ca2ccb84da9fb	CERTIFIED
M7	Master manifest over M1-M6	30e04e7b...	LOCKED
M8	$\text{rank}(H_{13}(L_w, J_0(143))) = g = 13$	e2d70821cd66588cd715dfe37a44122130f88d15584738f5f64a02ff7f7b0002	CERTIFIED
M8C	$Z=15$, $M^*=4/55$, 200 Hodge classes	02fe604876c3253ec61ce0a8b382c7b01a089d1d217ab200fc9975464a645323	CERTIFIED
M8J	$\delta=1.89\text{m}$; tidal<0.1g; OQs closed	298d440aae8ecc3808b413c7ce1b1cf19c92d359beb7664d837062e04b01b505	ARCH.CERT
M8K	FTL Morningstar; $B_M=21.768\text{MHz}$	0ae865a8812ce93b05461ec4483ad1714e24fc9be9de1e7bb54963da43592087	FTL.CERT
M8P	Logical clock; BSD rank=1; CONTACT ZERO	3e5f4f0432e6c4562f56f28aeb7a25a476df6b12d027601e038dce0d6f6ad6f6	CLOCK.CERT
M8Q	35/35 routes GREEN; 7-ABORT matrix	81e975cf...	SYS.CERT

Submitted to the Clay Mathematics Institute, Cambridge, MA | June 4, 2026

Full pipeline: 23-module causal DAG | All SHAs computed in environment, none fabricated

Source: github replit.dev Opera Numerorum | Stack: Python 3.12, mpmath 1.3.0, C gcc 80-bit, reportlab 4.5.1